

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to survive? Last week at the genome meeting here,* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using **computer analysis** to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes. The other team, which had previously required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those **predictions**

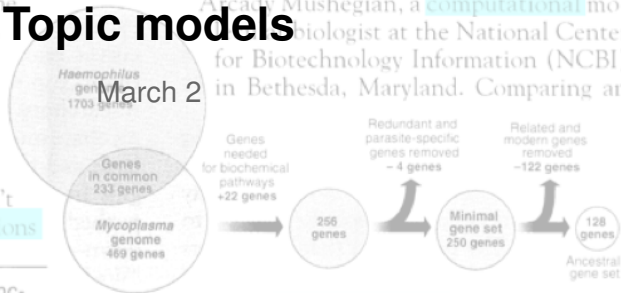
* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

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S&DS 265 / 565

Introductory Machine Learning

Example of Bayesian Inference: Topic models



Stripping down. **Computer analysis** yields an estimate of the minimum modern and ancient genes.

Yale

Quick notes

- Quiz 3 last week
- Midterm on Wednesday
- Will announce mid-semester grades after midterm
- Assn 3 due after mid-semester break

Quiz 3

Quiz Summary

Section Filter ▾

 Student Analysis

 Item Analysis

 Average Score

93%

 High Score

100%

 Low Score

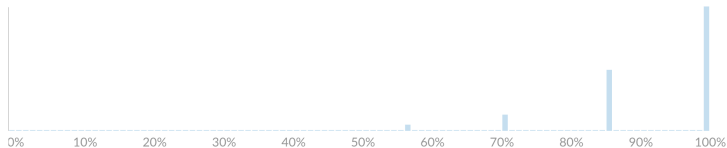
57%

 Standard
Deviation

0.74

 Average Time

12:04



Midterm review sessions

- ✓ Ruiqi: Friday, Feb 27, 7:00pm, KT 205
- ✓ Jianliang: Saturday, Feb 28, 5:00pm, TBA
- Dingyao: Monday, March 2, 8:00pm, KT 205
- John: Tuesday, March 3, 5:30pm, KT 205

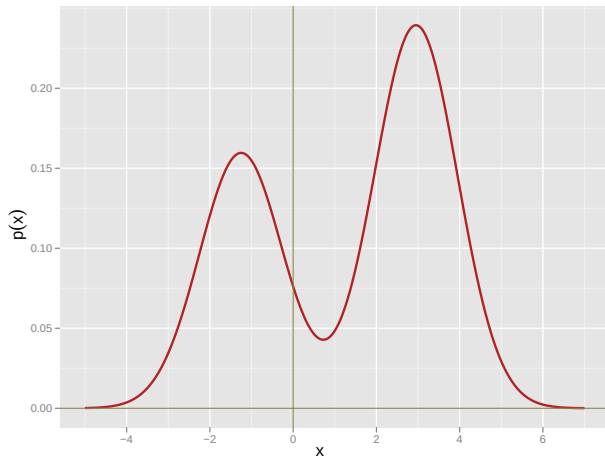
For Today

- Recap of Bayesian analysis
- Overview of topic models

Mixtures

- For each data point, we have a hidden (latent) variable: Which Gaussian generated the data point?
- $Z = 1$ it was generated from one model; $Z = 0$ it was generated from the other model
- We don't *observe* Z , it could be either value

Gaussian Mixture: All the Key Concepts



Bayesian Inference

The parameter θ of a model is viewed as a random variable.
Inference usually carried out as follows:

- Choose a *generative model* $p(x | \theta)$ for the data.
- Choose a *prior distribution* $\pi(\theta)$ that expresses beliefs about the parameter before seeing any data.
- After observing data $\mathcal{D}_n = \{x_1, \dots, x_n\}$, update beliefs and calculate the *posterior distribution* $p(\theta | \mathcal{D}_n)$.

Bayes' Theorem

The posterior distribution can be written as

$$p(\theta | x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n | \theta)\pi(\theta)}{p(x_1, \dots, x_n)} = \frac{\mathcal{L}_n(\theta)\pi(\theta)}{c_n} \propto \mathcal{L}_n(\theta)\pi(\theta)$$

where $\mathcal{L}_n(\theta)$ is the *likelihood function* and

$$c_n = p(x_1, \dots, x_n) = \int \mathcal{L}_n(\theta)\pi(\theta)d\theta$$

is the normalizing constant, which is also called *evidence*.

Important Example

Take model $X \sim \text{Bernoulli}(\theta)$.

This is a “coin flip”: $X = 1$ means “heads” and $X = 0$ means “tails.”

Natural prior is $\text{Beta}(\alpha, \beta)$ distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Important Example

Take model $X \sim \text{Bernoulli}(\theta)$.

Natural prior is $\text{Beta}(\alpha, \beta)$ distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Important Example

$X \sim \text{Bernoulli}(\theta)$ with data $\mathcal{D}_n = \{x_1, \dots, x_n\}$. Prior $\text{Beta}(\alpha, \beta)$ distribution

$$\pi_{\alpha, \beta}(\theta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

Let $s = \sum_{i=1}^n x_i$ be the number of “heads”

Posterior distribution $\theta \mid \mathcal{D}_n$ is another beta distribution!

Specifically, with

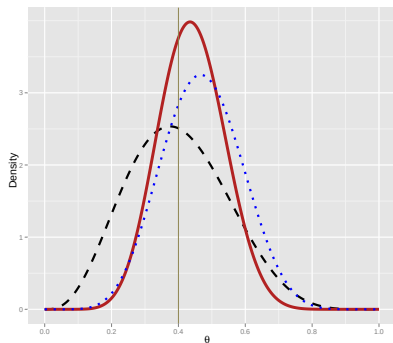
$$\tilde{\alpha} = \alpha + \text{number of heads} = \alpha + s$$

$$\tilde{\beta} = \beta + \text{number of tails} = \beta + n - s$$

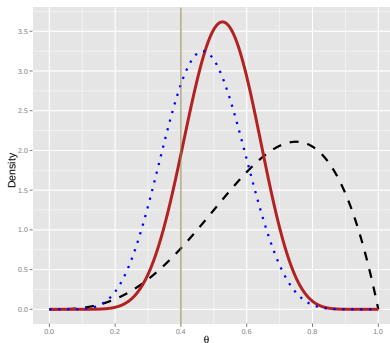
Showing this just uses the simple fact that $\theta^{\alpha-1} \theta^x = \theta^{x+\alpha-1}$

Example

$n = 15$ points sampled as $X \sim \text{Bernoulli}(\theta = 0.4)$, with $s = 7$ heads.



prior A



prior B

Prior distribution (black-dashed), likelihood function (blue-dotted),
posterior distribution (red-solid).

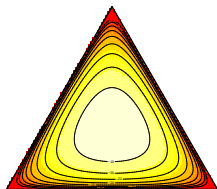
Dirichlet

Multinomial model with Dirichlet prior is generalization of the Bernoulli/Beta model.

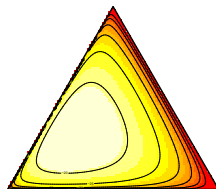
$$\text{Dirichlet}_{\alpha}(\theta) \propto \theta_1^{\alpha_1-1} \theta_2^{\alpha_2-1} \dots \theta_K^{\alpha_K-1}$$

where $\alpha = (\alpha_1, \dots, \alpha_K) \in \mathbb{R}_+^K$ is a non-negative vector.

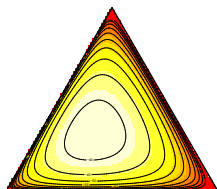
Example



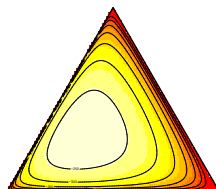
prior with Dirichlet(6,6,6)



likelihood function with $n = 20$



posterior distribution with $n = 20$



posterior distribution with $n = 200$

Example: Election Forecasting

<https://projects.economist.com/us-2020-forecast/president/how-this-works>



Today

Weekly edition

Menu



Forecasting the US elections

The Economist is analysing polling, economic and demographic data to predict America's elections in 2020

→ [Read more of our election coverage](#)

President

Senate

House

National forecast

How this works

COMPETITIVE STATES

Arizona

Florida

Georgia

Iowa

Michigan

Nevada

New Hampshire

North Carolina

Ohio

Pennsylvania

Texas

Wisconsin

ALL STATES

Alabama

How The Economist presidential forecast works

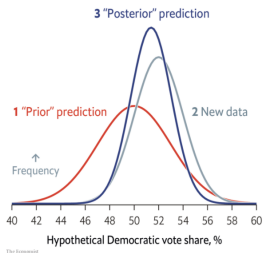
THIS YEAR, *The Economist* is publishing its first-ever statistical forecast of an American presidential election. Developed with the assistance of Andrew Gelman and Merlin Heidemanns, political scientists at Columbia University, our model calculates Joe Biden's and Donald Trump's probabilities of winning each individual state and the election overall. Its projections will be updated every day at <https://projects.economist.com/us-2020-forecast/president>.

In another first, we are [publishing the source code](#) for what we believe to be the most innovative section of the model. All readers are welcome to download it, explore how it works, tweak its parameters and run it

Example: Election Forecasting

<https://projects.economist.com/us-2020-forecast/president/how-this-works>

Three steps of Bayesian inference



Back to Bayes-ics

Readers acquainted with the workings of similar forecasting models may be surprised that the phrase “state polls” has not yet entered the equation. This exclusion is by design. Our model follows a logical structure first developed by Thomas Bayes, an 18th-century reverend whose ideas have shaped a large and growing family of statistical techniques. His approach works in two stages. First, before conducting a study, researchers

explicitly state what they believe to be true, and how confident they are in that belief. This is called a “prior”. Next, after acquiring data, they update this prior to reflect the new information—gaining more confidence if it confirms the prior, and generally becoming more uncertain if it refutes the prior (though not if the new numbers are so definitive that leave little room for doubt). In this framework, the expected distribution of potential vote shares in each state derived above is the prior, and state polls that trickle in during the course of the campaign are the new data. The result—a “posterior”, in Bayesian lingo—is our forecast.

Demo notebook

Demo code for Bayesian analysis

In this notebook we illustrate some of the basic models and priors for Bayesian inference. These concepts will be important for our discussions about "topic models."

First, we illustrate the situation where the parameter θ that we are modeling is a Bernoulli parameter. This can be thought of as the probability that flipping a certain coin comes up heads. The most commonly used prior distribution for this model is a beta distribution. Under a beta prior, the posterior distribution is again a beta distribution.

This is illustrated in the following simulation. As the sample size increases, the posterior distribution becomes concentrated on the true parameter. But as the variance of the prior decreases, the posterior distribution becomes more concentrated on the true parameter.

```
In [5]: import os, gzip
import numpy as np
import matplotlib.pyplot as plt
```

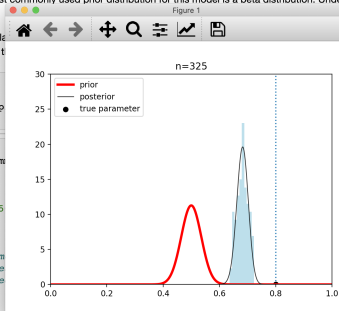
```
In [*]: %matplotlib qt
from scipy.special import gamma
from scipy import random
from scipy.stats import beta

theta = np.linspace(0,1,num=5)
fig = plt.figure(1)
plt.ion()
```

```
# The following are the parameters
# variance of the prior decreases as
# the posterior to be centered
```

```
scale = 100
a0 = scale*1
b0 = scale*1

sample_size = 100
```



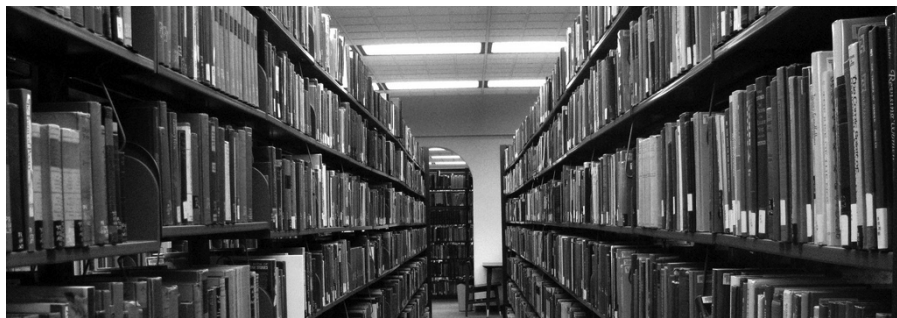
Summary

- Mixtures are latent variable models
- The mixing weight encodes a hidden variable
- Computing with mixtures uses basic probabilistic reasoning
- Bayesian inference is popular in ML
- In a Bayesian approach, the parameters are random, and the data are fixed.
- Three ingredients: Prior, likelihood, posterior

Example of Bayesian inference: Topic models

- Use of latent variables, mixtures
- More details and examples next week

Topic modeling



Topic modeling provides methods for automatically organizing, understanding, searching, and summarizing large electronic archives.

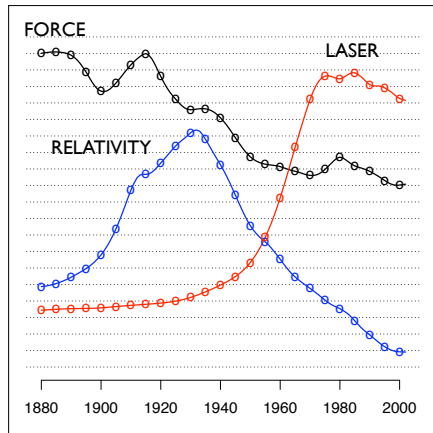
- ➊ Discover the hidden themes that pervade the collection.
- ➋ Annotate the documents according to those themes.
- ➌ Use annotations to organize, summarize, and search the texts.

Discover topics from a corpus

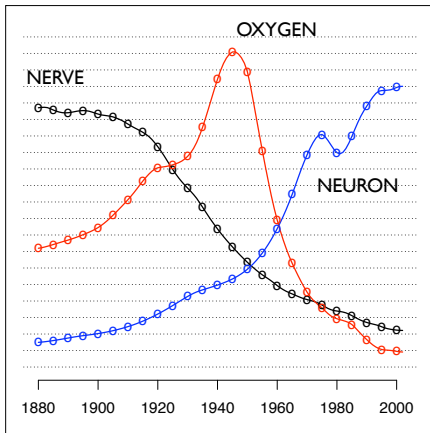
human	evolution	disease	computer
genome	evolutionary	host	models
dna	species	bacteria	information
genetic	organisms	diseases	data
genes	life	resistance	computers
sequence	origin	bacterial	system
gene	biology	new	network
molecular	groups	strains	systems
sequencing	phylogenetic	control	model
map	living	infectious	parallel
information	diversity	malaria	methods
genetics	group	parasite	networks
mapping	new	parasites	software
project	two	united	new
sequences	common	tuberculosis	simulations

Model the evolution of topics over time

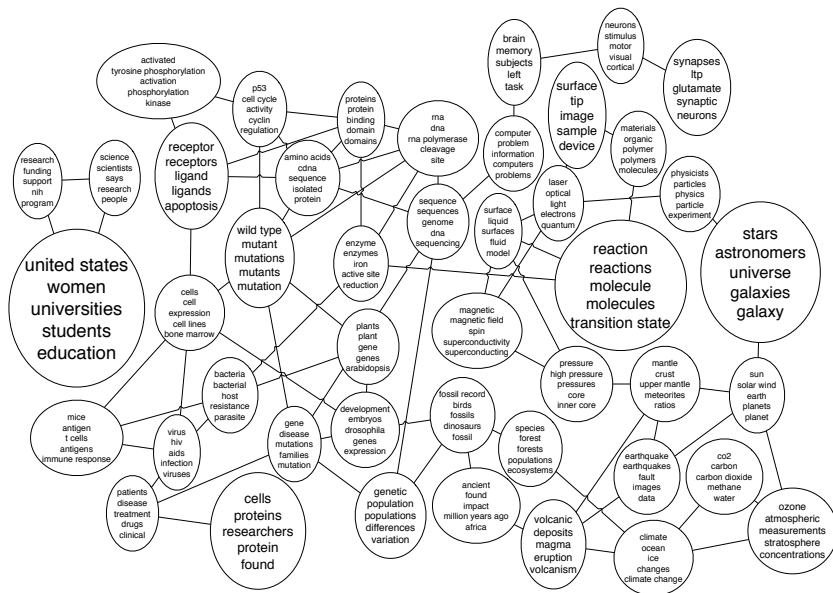
"Theoretical Physics"



"Neuroscience"



Model connections between topics



Probabilistic modeling

- ① Data are assumed to be observed from a generative probabilistic process that includes hidden variables.
 - *In text, the hidden variables are the thematic structure.*
- ② Infer the hidden structure using posterior inference
 - *What are the topics that describe this collection?*
- ③ Situate new data into the estimated model.
 - *How does a new document fit into the topic structure?*

Latent Dirichlet allocation (LDA)

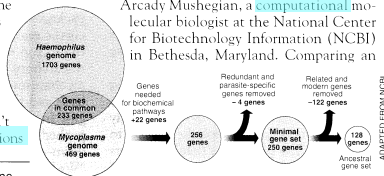
Seeking Life's Bare (Genetic) Necessities

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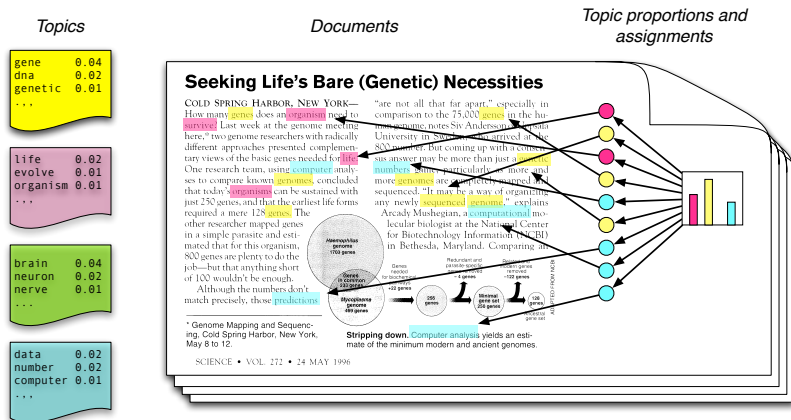
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Stripping down. Computer analysis yields an estimate of the minimum modern and ancient genomes.

SCIENCE • VOL. 272 • 24 MAY 1996

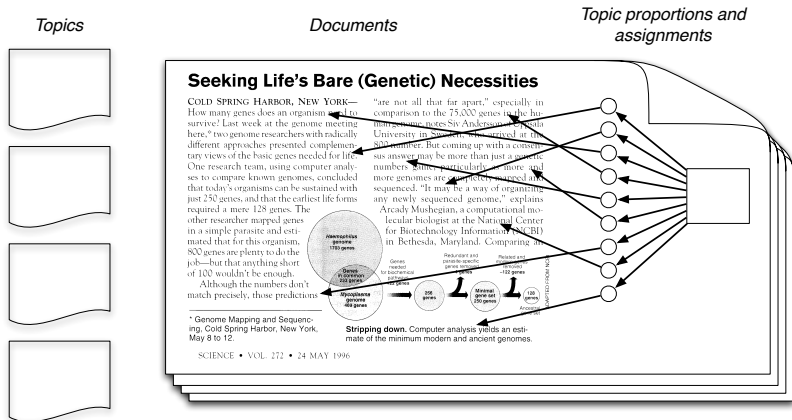
Simple intuition: Documents exhibit multiple topics.

Generative model for LDA



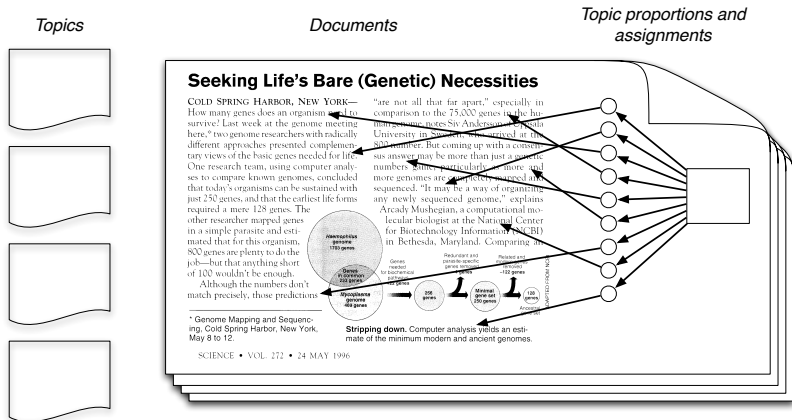
- Each **topic** is a distribution over words
- Each **document** is a mixture of corpus-wide topics
- Each **word** is drawn from one of those topics

The posterior distribution



- In reality, we only observe the documents
- The other structure are **hidden variables**

The posterior distribution



- Our goal is to **infer** the hidden variables
 - I.e., compute their distribution conditioned on the documents
- $p(\text{topics, proportions, assignments} \mid \text{documents})$

A little dicey



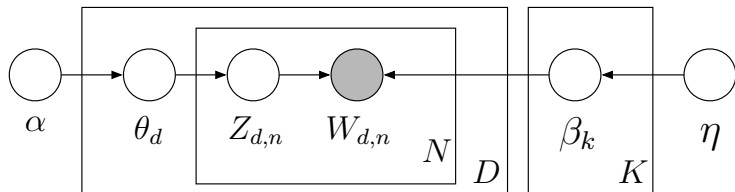
A little dicey



Left: Four topics β_k drawn from the prior; used across documents

Right: A distribution over topics θ ; used for a specific document

Example inference



- **Data:** The OCR'ed collection of *Science* from 1990–2000
 - 17K documents
 - 11M words
 - 20K unique terms (stop words and rare words removed)
- **Model:** 100-topic LDA model using variational inference.

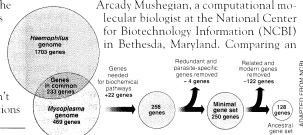
Example inference

Seeking Life's Bare (Genetic) Necessities

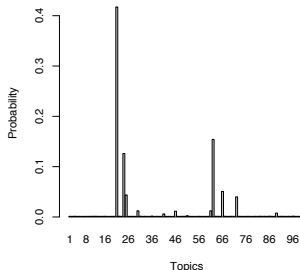
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Example inference

human	evolution	disease	computer
genome	evolutionary	host	models
dna	species	bacteria	information
genetic	organisms	diseases	data
genes	life	resistance	computers
sequence	origin	bacterial	system
gene	biology	new	network
molecular	groups	strains	systems
sequencing	phylogenetic	control	model
map	living	infectious	parallel
information	diversity	malaria	methods
genetics	group	parasite	networks
mapping	new	parasites	software
project	two	united	new
sequences	common	tuberculosis	simulations

Example inference (II)

Chaotic Beetles

Charles Godfray and Michael Hassell

Ecologists have known since the pioneering work of May in the mid-1970s (1) that the population dynamics of animals and plants can be exceedingly complex. This complexity arises from two sources: The tangled web of interactions that constitute any natural community provide a myriad of different pathways for species to interact, both directly and indirectly. And even in isolated populations the nonlinear feedback processes present in all natural populations can result in complex dynamic behavior. Natural populations can show persistent oscillatory dynamics and chaos, the latter characterized by extreme sensitivity to initial conditions. If such chaotic dynamics were common in nature, then this would have important ramifications for the management and conservation of natural resources. On page 389 of this issue, Costantino *et al.* (2) provide the most

convincing evidence to date of complex dynamics and chaos in a biological population—of the flour beetle, *Tribolium castaneum* (see figure).

It has proven extremely difficult to demonstrate complex dynamics in populations in the field. By its very nature, a chaotically fluctuating population will superficially resemble a stable or cyclic population buffeted by the normal random perturbations experienced by all species. Given a long enough time series, diagnostic tools from nonlinear mathematics can be used to identify the tell-tale signatures of chaos. In phase space, chaotic trajectories come to lie on "strange attractors," curious geometric objects with fractal structure and hence noninteger dimension. As they

move over the surface of the attractor, sets of adjacent trajectories are pulled apart, then stretched and folded, so that it becomes impossible to predict exact population densities into the future. The strength of the mixing that gives rise to the extreme sensitivity to initial conditions can be measured mathematically estimating the Liapunov expo-

nent, which is positive for chaotic dynamics and nonpositive otherwise. There have been many attempts to estimate attractor dimension and Liapunov exponents from time series data, and some candidate chaotic population have been identified (some insects, rodents, and most convincingly, human childhood diseases), but the statistical difficulties preclude any broad generalization (3).

An alternative approach is to parameterize population models with data from natural populations and then compare their predictions with the dynamics in the field. This technique has been gaining popularity in recent years, helped by statistical advances in parameter estimation. Good ex-



Cannibalism and chaos. The flour beetle, *Tribolium castaneum*, exhibits chaotic population dynamics when the amount of cannibalism is altered in a mathematical model.

The authors are in the Department of Biology, Imperial College at Silwood Park, Ascot, Berks, SL5 7PZ UK. E-mail: m.hassell@ic.ac.uk

Example inference (II)

problem	model	selection	species
problems	rate	male	forest
mathematical	constant	males	ecology
number	distribution	females	fish
new	time	sex	ecological
mathematics	number	species	conservation
university	size	female	diversity
two	values	evolution	population
first	value	populations	natural
numbers	average	population	ecosystems
work	rates	sexual	populations
time	data	behavior	endangered
mathematicians	density	evolutionary	tropical
chaos	measured	genetic	forests
chaotic	models	reproductive	ecosystem



Markov chain Monte Carlo

- Construct a **Markov chain** on the hidden variables, whose limiting distribution is the posterior.
- Collect **independent samples** from that distribution; approximate the posterior with them
- In **Gibbs sampling** the chain is defined by the conditional distribution of each hidden variable given observations and the current setting of the other hidden variables.

Idea behind Gibbs sampling

- Only the assignments $Z_{n,d}$ are needed
- From these we can infer the proportions θ_d (per document)
- And the topics β_k (per corpus)

Idea behind Gibbs sampling

At each time step in the algorithm, we have an assignment $Z_{n,d}$ of a topic to each word $w_{n,d}$ in every document d

Repeat forever:

- Select a word $w_{n,d}$
- Holding all of the other assignments $Z_{n',d'}$ fixed, calculate the probability distribution over $Z_{n,d}$ for that word
- Sample from that distribution to get a (potentially new) assignment $Z_{n,d}$

Extensions

Modeling richer assumptions

- Correlated topic model
- Dynamic topic model

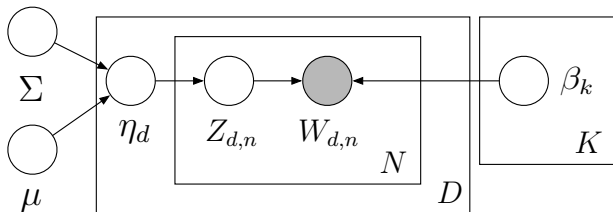
Shortcoming of the Dirichlet

- Dirichlet for topic proportions:

$$p(\theta | \alpha) \propto \prod_{j=1}^k \theta_j^{\alpha_j - 1}$$

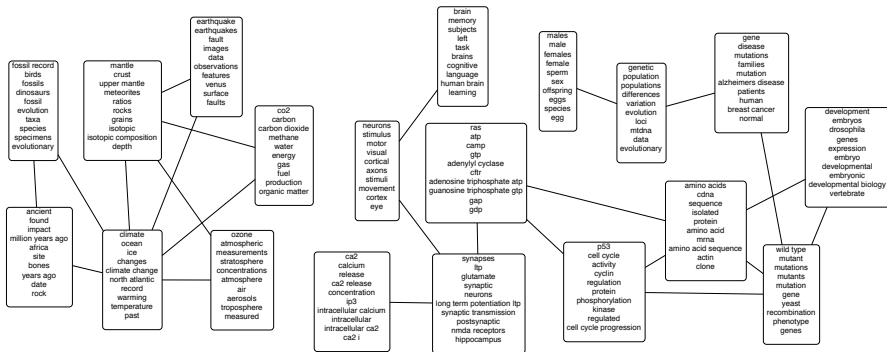
- Near independence of components makes it an unrealistic model
 - ▶ An article about *fossil fuels* is more likely to also be about *geology* than about *genetics*

Correlated topic model

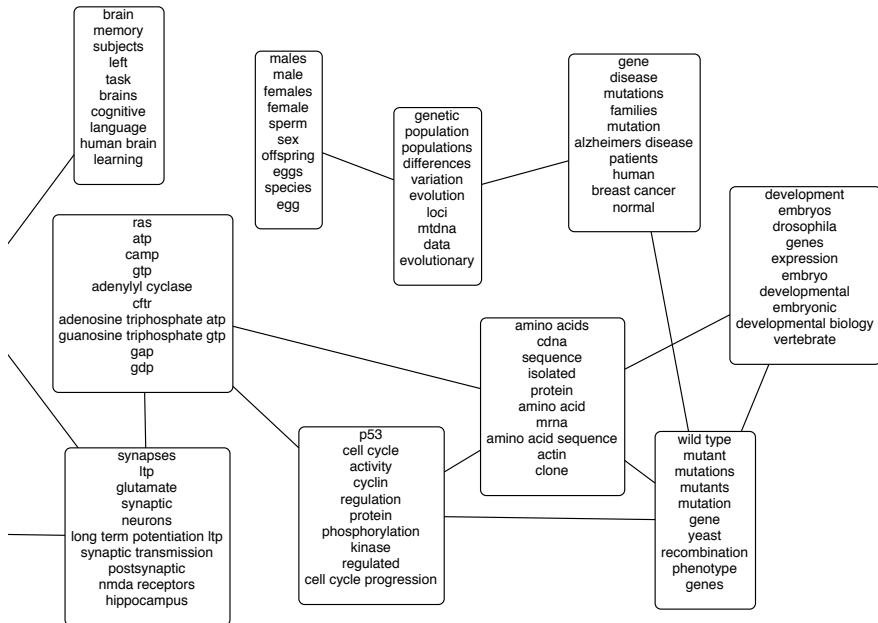


- Draw topic proportions from a logistic normal.
- Useful for:
 - ▶ providing a “map” of topics and how they are related;
 - ▶ better prediction via correlated topics.
- Sacrifice conjugacy: Posterior over θ does not have same form

Topic graphs



Topic graphs



Modeling Evolution of Topics

- In LDA, document order doesn't matter
- The topics should *evolve* over time
 - ▶ “Cleaning Birds” (1883)
 - ▶ “Interspecific Brood Parasitism in Blackbirds (Icterinae): A Phylogenetic Perspective” (1992)
- Many document collections have such dynamics: emails, query logs, news articles, etc.

Science 1893 ⇒ Science 1976

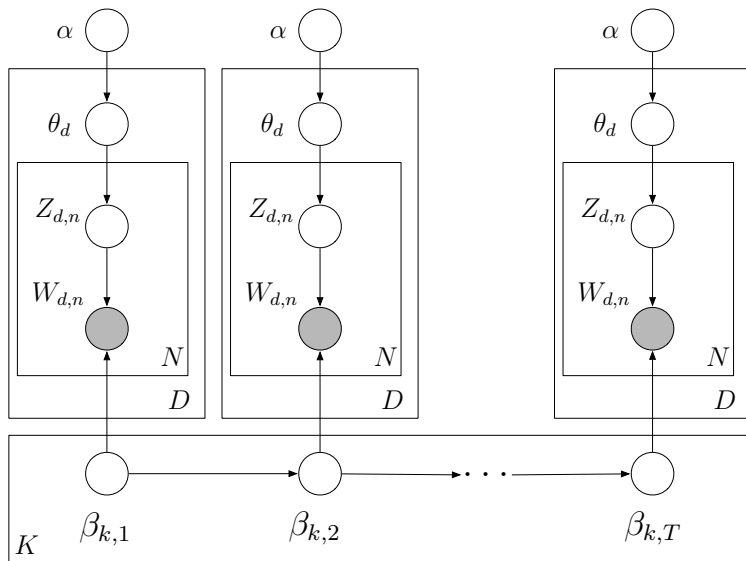
Administration	Limnology	Astronomy	Psychology
association	water	observatory	mind
meeting	lake	observations	nature
american	sea	stars	say
committee	waters	time	science
congress	lakes	made	psychology
members	gulf	astronomical	work
held	great	comet	knowledge
international	depth	star	truth
meetings	river	observed	religion
section	stream	telescope	human

Administration	Limnology	Astronomy	Psychology
house	water	mass	human
congress	concentrations	radio	attempts
science	mercury	objects	theory
bill	fish	astronomy	learning
nsf	samples	xray	ideas
president	soil	stars	new
budget	lake	astronomical	memory
office	ppm	sources	psychology
committee	concentration	observations	behavior
new	waters	observatory	complex

Time series topic model

1. Allow topics evolve between time slices
3. For each document in the current time slice:
 - a. Select a distribution over topics;
 - b. Generate the words from the resulting topic mixture.

Dynamic topic models



The Brain of the Orang (1880)

SCIENCE

in the Oyang, the parieto-occipital suture does not reach the calcarial, being separated from it by the "double-line pile de passage interne" of Grunhagen, or "anterior suture Schindlberger-Windling" of Blackhoff. I have noticed this separation as an anomaly more than once in man.

Gorilla, and seems to be unusual also in the Chimpanzee. In the female Chimpanzee, however, on the left side I found the parieto-occipital fissure passing into the calcarine, as in man. The frontal lobe is easily distinguished from the parietal by the fissure of Rolando, and

Chimpanzee. The number

[illegible]

Fig. 6.

* *Proceedings of the Academy of Neurologists*, Philadelphia.

According to Iliadoff, this disposition obtains in the Gorilla, and seems to be most abet in the Chimpanzee.

gained from the parallel of the history of Lebanon, and from the temporal by the House of Sybilus. In the Orang it is higher, wider, and more arched than in the Chinese. The exterior crest, considered in form,

Chimpanzee. The anterior central convolution is broad at the central fissure runs into the post-central convolution above and below, as in man. It is difficult, however, to identify the distal frontal convolution in man and

to identify the three frontal convolutions seen in man and the chimpanzee, the frontal lobe of the Orang dividing surface into two convolutions, the middle one being badly defined. This is also consistent with the results of the present study.

defined. This is due somewhat to the length of the pre-cerebral fissure, which is as long as the fissure of Rolando, extending farther upward than in man. There was

working particularly noticeable along the base of the frontal lobe; on the medial surface it runs into the parietal. The part above the callosal-marginal fissure is

The Ovaries are not as distinctly divided into convolutions as in man, though there are not constantly present even in all human ovaries. The parietal lobe is separated from

the frontal by the central fissure, from the occipital and temporal incompletely, by the parieto-occipital and Sylvian fissures. The posterior-central convolution is

well defined. The parietal feature in the Chang is more striking than that of man, resembling the Gorilla's; it is twice as long as the corresponding feature in the Chin

range as long as the corresponding bones in the Campanian, extending from the transverse occipital foramen, as is sometimes the case in man, almost to the base of the foramen of Salsola. It is subcylindrical and without a break and

divides the parasternal lobe completely into upper and lower parasternal lobules. The upper parasternal lobule is bounded posteriorly by the vertical fissure. *Species:* *contortus* B. is com-

internally by the parietal fissure; posteriorly it is separated from the occipital lobe, internally by the parieto-



Fig. 6

occipital fissure; externally it is continuous with the occipital lobe, as the first occipital gyrus, internally it is separated from the posterior cerebral commissure, more

separated from the posterior central convolution more completely than with the central fissure. There is in the Oryz. also, a

Incisor running parallel with the parietal, which subdivides the upper parietal lobule into lesser and outer portions. The pterion, or the square on the middle

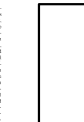
side of the parietal lobe between the parieto-occipital

A histogram showing the distribution of the number of non-zero elements in the matrix A. The x-axis is labeled 'Number of non-zero elements' and ranges from 0 to 1000. The y-axis is labeled 'Frequency' and ranges from 0 to 100. The distribution is highly skewed, with a peak frequency of approximately 100 at around 500 non-zero elements.

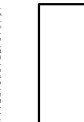
Representation of the Visual Field on the Medial Wall of Occipital-Parietal Cortex in the Owl Monkey (1976)

what is seen in Fig. 2, which illustrates the organization of the other cortical visual areas that have been mapped in the owl monkey, the border between the medial area and the second visual area corresponds to a peripheral portion of the horizontal meridian. In other experiments in the dorsomedial area, we found that receptive fields located near its common border with the medial area began near the vertical meridian and in the lower quadrant and proceeded in a broad loop in the periphery toward the horizontal meridian (Figs. 2, 5). Thus, as is shown in Figs. 1 and 2, the common border between the dorsomedial and the medial areas corresponds in part to the lower field vertical meridian and the peripheral portions of the lower visual quadrant. Possibly, the medial area is adjacent to por-

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Exploring the UN General Debates with Dynamic Topic Models



Luke Lefebure

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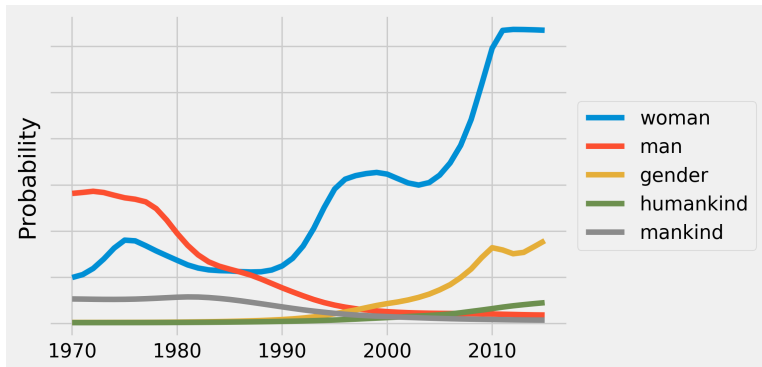
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exploring-the-un-general-debates-with-dynamic-topic-models-72dc0e307696

Human rights

	1970	1980	1990	2000	2010
0	right	right	right	right	right
1	human	human	human	human	human
2	people	people	freedom	law	law
3	international	freedom	people	democracy	woman
4	principle	international	democracy	respect	freedom
5	justice	political	respect	international	respect
6	freedom	principle	law	people	people
7	law	respect	international	freedom	democracy
8	state	justice	principle	principle	rule
9	must	social	state	must	international

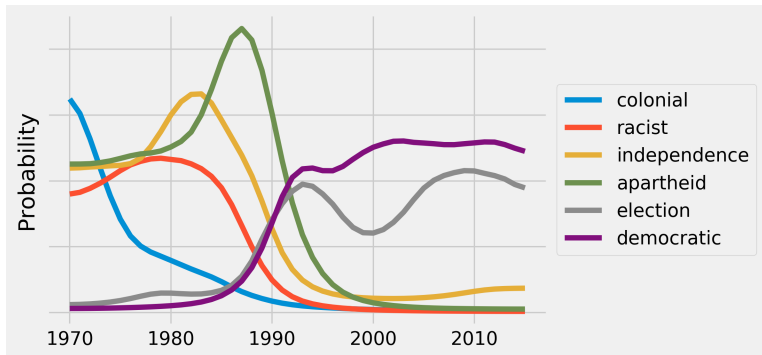
Human rights



Apartheid

	1970	1980	1990	2000	2010
0	africa	africa	africa	african	african
1	african	south	south	peace	country
2	south	african	african	africa	government
3	colonial	namibia	apartheid	country	africa
4	people	people	people	government	people
5	regime	regime	government	community	peace
6	southern	independence	country	democratic	political
7	government	apartheid	namibia	international	community
8	territory	racist	community	republic	democratic
9	apartheid	southern	process	effort	national

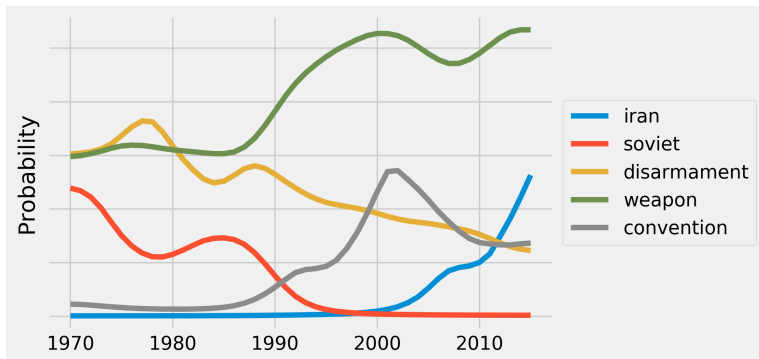
Apartheid



Nuclear arms

	1970	1980	1990	2000	2010
0	nuclear	nuclear	nuclear	weapon	nuclear
1	disarmament	disarmament	weapon	nuclear	weapon
2	weapon	weapon	disarmament	convention	non
3	soviet	arm	treaty	arm	proliferation
4	arm	state	arm	treaty	arm
5	treaty	race	state	disarmament	treaty
6	union	military	chemical	proliferation	international
7	agreement	treaty	agreement	international	disarmament
8	power	soviet	proliferation	non	convention
9	state	power	soviet	destruction	state

Nuclear arms



Topic modeling for equations

Topic	Generated Equations
Quantum physics	• $E = \hbar \frac{\partial^2 S}{\partial t^2} \left(\frac{\partial \varphi}{\partial c} \right) - \frac{k}{\hbar^2} \frac{\partial B}{\partial t} (t + \partial_t \delta).$ • $\Psi_{\text{pr}} = \sum_{\mathbf{l}} (\psi_{\mathbf{r}+\uparrow} - \psi_{\mathbf{r}\downarrow}^\dagger) + \sum_{\mathbf{r}'} (\psi_{\mathbf{r}\downarrow,\uparrow}^\dagger - \psi_{\mathbf{r}\downarrow} \sigma^\dagger).$
Particle physics	• $\mathcal{H} = \frac{2}{4}(\partial_\mu \phi)^2 + 2m\phi_\nu(\phi) + \frac{1}{2}m^2(\phi)(1 - \phi^2)^2.$ • $m_{\text{eff}}(M) = 1.4 \cdot 10^{-13} \text{ GeV}.$
Relativity	• $\mathcal{M} = \frac{2}{2}g^{\mu\nu}(f_{\mu\nu,\mu} - g_{\mu\nu,\nu} + g_{\nu\nu,b}f_{\mu,\nu}) + \frac{1}{2}g^{\mu\nu}.$ • $T_{\mu\nu} = \int_0^\infty ds_{\mu\nu} ds^2 + a_\mu^2 dr^2 + r^2 d\Omega^2.$

Uber Topics

From a former student:

Hi Prof. Lafferty,

I took your course last spring. I just wanted to let you know that I'm currently topic modeling at my current job at Uber!

We're using topic models at Uber to discover topics in rider feedback – when riders write comments about their driver after the trip. We're trying to find topics such as 'unprofessional driver', 'driver no-show', 'sexual harassment', etc.

Summary

- Topic models automatically extract “semantic themes” from large document collections
- Use mixtures and latent variables
- Estimating Bayesian posterior done with Gibbs sampling
- Many extensions are possible